

Clinical Data Visualizer – User Guide

Clinical Data Visualizer Team

April 22, 2026

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1 Introduction

Clinical Data Visualizer is an interactive dashboard for visualizing clinical physiological signals. It allows clinicians and researchers to explore, compare, and annotate time-series data from multiple medical devices in a single unified interface.

1.1 Key Features

- **Multi-source visualization:** Display signals from multiple clinical data sources simultaneously (Philips, FluxMed, Mindray, EIT, Servo-U, syringe pumps, plus a generic “Other” source for any CSV/parquet).
- **Generic “Other” data source:** Drop any CSV or Parquet file with a datetime column into an `other/` folder — signals are auto-discovered and can be configured per file.
- **Interactive plots:** Zoom, pan, and explore data at any time scale with automatic resampling.
- **Data inspection:** Preview available columns, point counts, and time ranges for every source before running the full visualization — exportable to CSV.
- **Annotations:** Draw lines and rectangles directly on plots to mark events or regions of interest. Annotations are saved and persist across sessions.
- **Flexible configuration:** Choose which signals to display, customize labels, units, colors, and group related signals together — in JSON or Excel format.
- **Cross-datasource phase loops:** Define loop entries to build phase plots that combine signals from different devices.
- **Per-datasource timezone control:** Override the timezone of any supported source via `additional_information.timezone` (all plots still render in the configured display timezone).
- **Export:** Generate standalone HTML visualizations for sharing.

The full list of supported data sources — with folder keywords, accepted file extensions, and typical signals — lives in Section 3: **Patient Data & Supported Data Sources**.

2 Launching the Application

2.1 Starting the App

Locate the **ClinicalDataVisualizer** executable in the application folder and double-click it.

A terminal window will appear showing the application starting up. After a few seconds, your default web browser will automatically open at:

`http://127.0.0.1:8050`

If the browser does not open automatically, manually navigate to the address above.

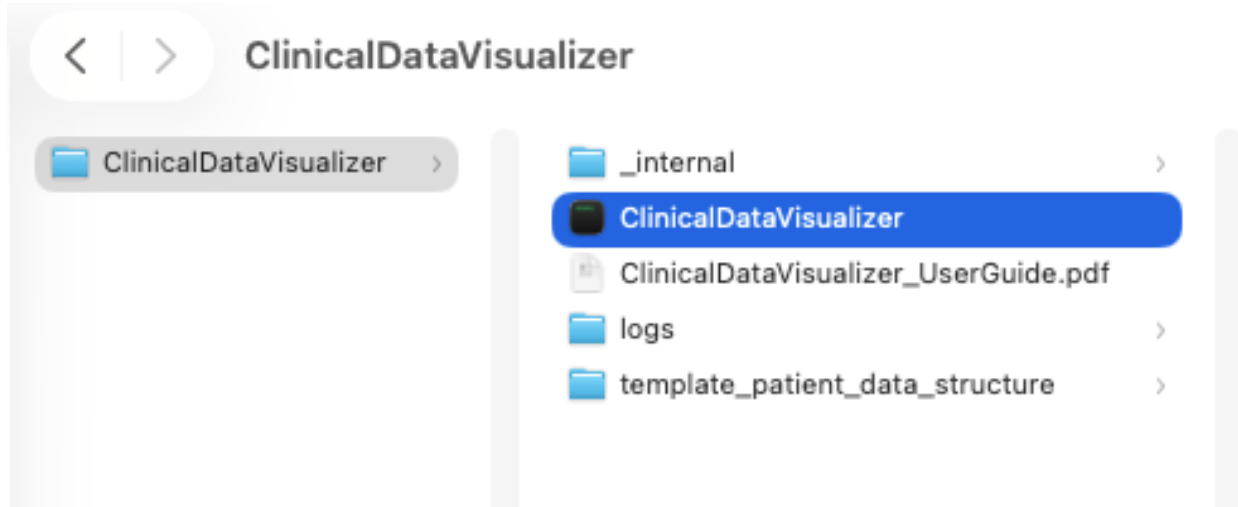


Figure 1: Application launch screen

The app also comes with the userguide (this document) and template folder to help organizing patient data for the app. Once runned for the first time, a **logs** folder will also appear, helping to debug if needed.

2.2 Application Overview

The interface is organized top-to-bottom in the following order:

1. **Database Options** – Three buttons side-by-side:
 - **Upload config file** (blue) – load a custom **database options** file, either **.json** or **.xlsx**.
 - **Reload last config** (grey) – appears only if a previously uploaded config was cached; restores it with one click.
 - **Default visualization (all sources)** (green) – enables every registered data source with default settings, no file needed.
2. **Patient Options** – Configure data folder, time range, and per-source settings. The form is generated dynamically from the loaded database options.
3. **Action buttons** – **Process visualization** (orange) and **Inspect data** (teal).
4. **Annotations Controls** – Shape dropdown + Modify/Delete buttons (visible after a successful visualization).
5. **Inspection pop-up** – Full-screen overlay triggered by the Inspect button, with per-source status badges, column tables, and a CSV download.
6. **Visualization Area** – Interactive plots.

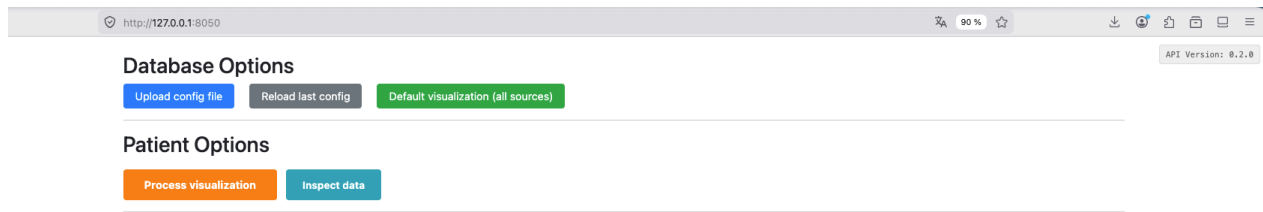


Figure 2: Application main interface

3 Patient Data & Supported Data Sources

Each patient’s data must be organized in a root folder with **one subfolder per data source**. The application automatically identifies data sources based on keywords in subfolder names.

You only need subfolders for the data sources you actually have — empty or missing subfolders are silently skipped. The `cdv_visu/` subfolder is created automatically the first time you process a patient. It contains annotations, `.html` visualizations, formatted data.

3.1 Example Folder Layouts

A full setup with several devices:

```
Patient1/
  philips_waves/
  philips_numerics/
  eit/
  fluxmed_signals/
  fluxmed_parameters/
  servo_u/
  syringe/
  cdv_visu/          ← auto-created,
```

A minimal setup:

```
Patient1/
  philips_waves/
  cdv_visu/
```

An “**Other**”-heavy setup — drop any CSV/Parquet file with a datetime column into `other/`; each file is configured independently via `other::<stem>` keys in the database options (see [dedicated section](#) below):

```
Patient1/
  other/
    waves.parquet      → configured under "other::waves"
    numerics.csv       → configured under "other::numerics"
    syringe_log.csv    → configured under "other::syringe_log"
  cdv_visu/
```

3.2 Folder Naming Rules

Folder names are **flexible** — they just need to contain the required keywords:

- **Case-insensitive** — `Philips_Waves`, `PHILIPS-WAVES`, `philips waves` all match.
- **Any separator** — underscore, dash, space, or none.
- **Any order** — `waves_philips` works just as well as `philips_waves`.
- **Partial keywords don’t match** — `flux` alone will not match `fluxmed_*`; the full word is required.

A few sources are also **identified by file extension** when the folder is ambiguous or generic (e.g., `.asc` files inside a folder are enough to classify it as EIT, `.sta` for Servo-U, `.xml` for Mindray Scope). See the canonical table below for exact keywords and accepted extensions per source.

3.3 Canonical Data Source Table

The table below is the single source of truth for supported data sources and is what all folder-discovery / skill logic refers to.

Source	Module name	Folder keywords	Accepted extensions (ordered by preference)	Discovery mode	Typical signals
Philips Waves	philips_waves	philips, waves	.parquet, .csv	Single file	ART, PAP, CO2, respiratory pres-sure/volume
Philips Numerics	philips_numerics	philips, numerics	.parquet, .csv	Single file	Heart rate, SpO2, FiO2, blood pressure
EIT (PulmoVista)	eit	eit	.asc	All files	Global/local impedance, impedance percentages
FluxMed Signals	fluxmed_signals	fluxmed, signals	.parquet, .txt, .csv	Single file	Respiratory waveforms
FluxMed Parameters	fluxmed_parameters	fluxmed, parameters	.parquet, .txt, .csv	Single file	Respiratory parameters
Servo-U	servo_u	servo	.sta	All files	Ventilator waveforms and settings
Mindray Scope	mindray_scope	mindray	.xml, .csv	All files	Monitor waveforms (ECG, SpO2, pressure)
Mindray Respi Waves	mindray_respi_waves	mindray, resp, wave	.parquet, .csv	Single file	High-frequency respiratory waveforms
Mindray Respi Numerics	mindray_respi_numerics	mindray, resp, numeric	.parquet, .csv	Single file	Respiratory parameters (Vt, RR, PEEP, etc.)
Syringe	syringe	syringe	.parquet, .csv	Single file	Infusion rates and volumes
Other (Generic)	other	other	.csv, .parquet	All files (one entry per file)	Any time-series with a datetime column

Single file sources expect exactly one data file per folder. When several formats coexist (e.g., `data.csv` and `data.parquet`), the most preferred extension wins. If multiple unrelated stems remain after that filter, the source is skipped and a warning is logged.

All files sources load every matching file in the folder and concatenate them. The **Other** source is special: each file produces an independent entry named `other::<stem>` (see below).

Per-source configuration options (`field_display`, `signals`, `grouped_fields`, `loop`, `additional_informations`, etc.) are documented in the [Configuration File Reference section](#).

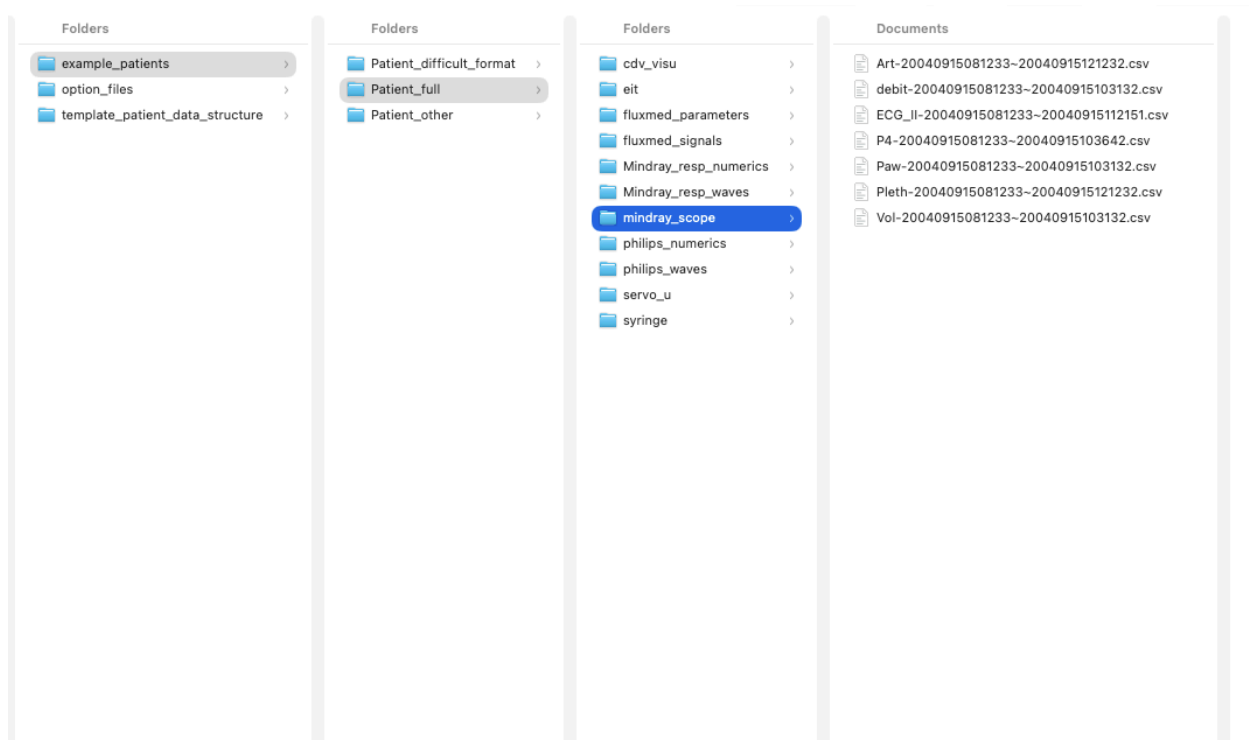


Figure 3: Patient folder structure example

3.4 Generic “Other” Data Source

other is the **generic escape hatch** for any CSV or Parquet file that has a datetime column but does not fit any of the specialised sources. It is the most natural entry point for users who already have well formatted data.

How it works:

- Drop any number of `.csv` or `.parquet` files into an `other/` subfolder.
- Every file is discovered automatically — no separate folder or config key per file.
- Each file produces an **independent entry** keyed by its stem (filename without extension): a file named `waves.parquet` becomes `other::waves`, `numerics.csv` becomes `other::numerics`, and so on.
- Columns within a file are exposed as `<stem>::<column_name>` so names stay globally unique (important for cross-datasource groups and loops).

Datetime column auto-detection. The loader tries the common names first (`datetime`, `timestamp`, `time`, `date`, `date_time` — case-insensitive). If none match, it falls back to scanning non-numeric columns for parseable timestamps.

Per-file configuration. In `database_options`, use `other::<stem>` keys — one block per file — exactly like any other datasource:

```
"other::waves": {
  "field_display": ["art", "pleth"],
  "signals": {
    "art": { "label": "Arterial Pressure", "unit": "mmHg", "color": "red" },
    "pleth": { "label": "Plethysmography", "unit": "AU", "color": "purple" }
  },
  "additional_informations": { "timezone": "Europe/Paris" }
},
```

```
"other::numerics": {  
  "field_display": ["FC", "SpO2"],  
  "additional_informations": { "timezone": "UTC" }  
}
```

Per-file timezone. Each `other::<stem>` block may declare its own `additional_informations.timezone` — useful when the CSV you dropped in comes from a device in a different timezone than your default.

Inspection shows one entry per file. When you click *Inspect data* for a patient with an `other/` folder, the modal lists one row per file (`other::waves`, `other::numerics`, ...), each with its own date range and column list, so you can verify that every file was correctly discovered and parsed.

See `example/option_files/example_database_options_other.json` in the source repository for a full reference configuration.

4 Loading Database Options

Database options define **which data sources to enable** and **how signals should be displayed** (labels, units, colors, grouping).

There are three ways to get a database options config into the app, presented here in the order you are most likely to use them.

4.1 Option 1: Upload a Custom Configuration File

Click the blue “**Upload config file**” button to load a custom configuration. Two formats are accepted:

- **.json extension** — a JSON file following the structure described in [Configuration File Reference section](#).
- **.xlsx extension** — an Excel spreadsheet following the column layout documented in the same reference. The spreadsheet is converted to the equivalent JSON structure on load.

This gives you full control over which sources are enabled and how each signal is displayed.

When the upload succeeds the file is **cached locally** at `~/.clinical_data_visualizer/last_database_options.json` — this is what powers Option 2 below. Such a database option file should only contains signal metadata (labels, colors, units, field mappings) and no patient data or PHI, to ensure no sensitive information is cached and so leaves the original patient folder.

4.2 Option 2: Reload Last Config (Daily Workflow)

If a custom configuration was previously uploaded, a grey “**Reload last config**” button appears automatically on startup. Click it to instantly restore the last used configuration without browsing for files — ideal when you re-open the app every day with the same setup.

This button is **hidden** when no cached configuration exists.

4.3 Option 3: Default Visualization (Quick Start)

Click the green “**Default visualization (all sources)**” button. This automatically enables **every registered data source** with its built-in default display settings — no configuration file needed. This is the recommended starting point for new users, and it automatically picks up any new data sources added to the library without requiring a config update. Note that if you have hundreds of timeseries, they will all be plot and you may experience performance issues on the app.

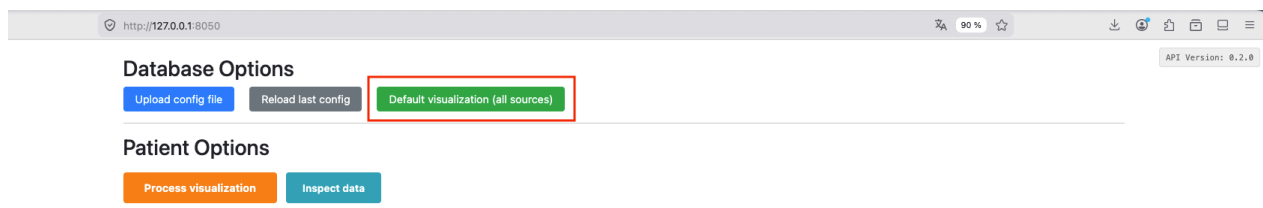


Figure 4: Default visualization button

5 Configuring Patient Options

After loading database options, the **Patient Options** form appears. It is divided into two parts: global options and per-source options.

5.1 Global Options

These apply to all data sources:

Option	Description
Path to data (folder)	Full path to the patient's root data folder
Time start filter	Start of the time window to display (format: YYYY-MM-DD HH:MM:SS). Leave empty to use all available data.
Time end filter	End of the time window to display. Leave empty to use all available data.
Re-use data if already loaded once	When checked, reuses previously cached <code>.parquet</code> files from the <code>cdv_visu/</code> folder, significantly speeding up subsequent loads. Un-tick if raw patient data has been modified

The screenshot shows a web browser window with the URL `http://127.0.0.1:8050`. The page title is 'Database Options'. Below the title are three buttons: 'Upload config file', 'Reload last config', and 'Default visualization (all sources)'. A green message bar says 'Successfully loaded example_database_options.xlsx'. The main section is 'Patient Options', which is divided into 'Global Patient Options' and 'Specific Options'. The 'Global Patient Options' section has a red border and contains the following fields: 'Path to data (folder)' (text input), 'Time start filter' (text input with format 'YYYY-MM-DD HH:MM:SS'), 'Time end filter' (text input with format 'YYYY-MM-DD HH:MM:SS'), and 'Re-use data if already loaded once' (checkbox, checked). The 'Specific Options' section contains four cards: 'Philips scope - waves' (Time shift: - 0 +), 'EIT - PulmoVista' (Time shift: - 0 +, Day of EIT recording: YYYY-MM-DD HH:MM:SS), 'Philips scope - numerics' (Time shift: - 0 +), and 'Syringe' (Time shift: - 0 +). At the bottom are two buttons: 'Process visualization' (orange) and 'Inspect data' (blue).

Figure 5: Global patient options

5.2 Per-Source Options

Below the global options, each enabled data source may have additional settings displayed in individual cards arranged in a two-column grid. Common per-source options include:

- **Time shift** (seconds): Adjust the time alignment of a source relative to others. Useful when devices were not perfectly synchronized.
- **Day**: Specify the recording date for sources that require it (e.g., EIT data).

Only data sources present in the loaded database options will show their configuration cards.

The screenshot shows a web browser at `http://127.0.0.1:8050` displaying the 'Database Options' page. The page has a top navigation bar with a status 'API Version: 0.2.0'. Below the title, there are three buttons: 'Upload config file' (blue), 'Reload last config' (grey), and 'Default visualization (all sources)' (green). A green message states 'Successfully loaded example_database_options.xlsx'.

The 'Patient Options' section is titled 'Global Patient Options' and contains a form with the following fields:

- 'Path to data (folder)' with a text input 'Path...'.
- 'Time start filter' with a date-time input 'YYYY-MM-DD HH:MM:SS'.
- 'Time end filter' with a date-time input 'YYYY-MM-DD HH:MM:SS'.
- 'Re-use data if already loaded once' with a checked checkbox.

The 'Specific Options' section, highlighted with a red border, contains four configuration cards:

- Philips scope - waves**: 'Time shift' input with a value of 0 and minus/plus buttons.
- EIT - PulmoVista**: 'Time shift' input with a value of 0 and minus/plus buttons, and 'Day of EIT recording' input with a date-time format 'YYYY-MM-DD HH:MM:SS'.
- Philips scope - numerics**: 'Time shift' input with a value of 0 and minus/plus buttons.
- Syringe**: 'Time shift' input with a value of 0 and minus/plus buttons.

At the bottom of the 'Specific Options' section are two buttons: 'Process visualization' (orange) and 'Inspect data' (teal).

Figure 6: Per-source options

6 Processing Data

Once patient options are configured, two actions are available from the action row. **Both run the same data pipeline** — find → load → format — and share the same per-datasource progress bar. What differs is only the outcome: the orange button builds interactive plots, the teal button produces a structured summary of what was found.

Recommended workflow: start with **Inspect data** (teal button). It completes faster because no plots are built, and it immediately surfaces loading errors, missing folders, or unexpected column names.

6.1 Process Visualization (orange button)

Click the large orange “**Process visualization**” button to generate the plots. The steps are:

1. **Validation:** The application verifies that all mandatory fields are filled in and that the data folder exists.
2. **Data Discovery:** For each enabled data source, the application scans the patient folder for matching subfolders and files.
3. **Data Loading:** Raw data files are parsed according to each source’s format.
4. **Formatting:** Signals are filtered, resampled, and converted using your database options (labels, units, time range).
5. **Caching:** Processed data is saved as `.parquet` files in the `cdv_visu/` subfolder for faster reloading next time. Tick “**Re-use data if already loaded once**” (`quick_load`) to skip raw parsing on subsequent runs and read the cache instead.
6. **Plot Generation:** Interactive Plotly figures are created and displayed in the visualization area.

While processing runs, a **per-datasource progress bar** appears below the action row. It shows (`completed / total`): `<datasource>` and its color matches the action — orange for visualization. The bar advances as each datasource completes; it never reaches 100% while the last source is still being processed (by design, so you always see which source is active).

A success message appears when processing completes. If no data is found for a source, it is silently skipped.

6.2 Inspect Data (teal button)

The teal “**Inspect data**” button runs the same `find → load → format` pipeline but **stops before signal extraction and plot building**. It is the **recommended first step** when opening a new patient folder: no figures are built so it is significantly lighter, and it immediately exposes any loading errors, missing folders, or unexpected column names — letting you catch problems before running the heavier visualization.

The same progress bar is shown during inspection, colored teal.

When inspection completes, a **full-screen inspection modal** opens, with one section per datasource containing:

- **Status badge** — colored per outcome:
 - green `ok` — data loaded successfully
 - orange `file_not_found` — no matching file/folder in the patient directory
 - red `load_error` or `format_error` — loading or formatting raised an exception (the error message is shown below the badge)
- **File path** — the detected data file or folder.
- **Date ranges** — both the raw (unfiltered) and filtered time ranges.
- **Columns table** — each column with: raw name, whether it is configured in the database options, raw point count, filtered point count, and first/last filtered timestamps.

Note for the Other (Generic) source: because the `other` folder may contain several independent files, the inspection modal shows **one entry per file** (e.g. `other::waves`, `other::numerics`) rather than a single aggregated entry. Each entry has its own date range and column list.

A “Download CSV” button in the modal header exports the full inspection result as a CSV for offline analysis, sharing, or import into the `generate-database-options` helper.

Database Options

Upload config file | Reload last config | Default visualization (all sources)

Data Inspection Download CSV Close

philips_waves ok

File: /Users/alexis/ClinicalData/visualizer/example/example_patients/Patient_full/philips_waves/data_waves_anonymized.parquet
 Date range in file: 04-09-15 08:12:33 → 04-09-15 08:13:28
 After filter: 04-09-15 10:12:38 CEST → 04-09-15 10:13:33 CEST

Column	Configured	Raw pts	Filtered pts	% retained	First (filtered)	Last (filtered)
art	×	10,000	10,000	100.0%	04-09-15 10:12:38 CEST	04-09-15 10:13:33 CEST
vol	×	0	0	—	—	—
pleth	×	10,000	10,000	100.0%	04-09-15 10:12:38 CEST	04-09-15 10:13:33 CEST
paw	×	0	0	—	—	—
p4	×	0	0	—	—	—
debit	×	0	0	—	—	—

eit ok

File: /Users/alexis/ClinicalData/visualizer/example/example_patients/Patient_full/cdv_visu/eit.parquet
 Date range in file: 0.5651115972 → 0.5718922685
 After filter: 15-04-09 13:33:45 CEST → 15-04-09 13:43:31 CEST

Column	Configured	Raw pts	Filtered pts	% retained	First (filtered)	Last (filtered)
Image	×	11,718	11,718	100.0%	15-04-09 13:33:45 CEST	15-04-09 13:43:31 CEST
Global	✓	11,718	11,718	100.0%	15-04-09 13:33:45 CEST	15-04-09 13:43:31 CEST
Local 1 (X:11 Y:22)	×	11,718	11,718	100.0%	15-04-09 13:33:45 CEST	15-04-09 13:43:31 CEST
Local 2 (X:21 Y:22)	×	11,718	11,718	100.0%	15-04-09 13:33:45 CEST	15-04-09 13:43:31 CEST
Local 3 (X:11 Y:11)	×	11,718	11,718	100.0%	15-04-09 13:33:45 CEST	15-04-09 13:43:31 CEST

Figure 7: Inspect data pop up

6.3 Inspecting from the Command Line

The same inspection is available as a standalone script:

```
python scripts/inspect_patient_data.py <patient_folder> \
  [--database-options <path>] \
  [--patient-options <path>] \
  [--output-csv out.csv] \
  [--verbose]
```

Without `--database-options`, all registered datasources are inspected with their defaults. Pass `--output-csv` to save the same per-column table the UI download produces.

7 Interacting with Plots

7.1 Navigation Controls

Each plot provides a toolbar (top-right corner) with the following tools:

Tool	Action
Zoom	Click and drag to zoom into a rectangular region
Pan	Click and drag to move the view
Zoom In / Zoom Out	Incremental zoom buttons
Autoscale	Reset the view to fit all data
Reset Axes	Return to the original view
Download as PNG	Save the current plot view as an image

7.2 Zooming and Panning

- **Scroll wheel:** Zoom in/out on the x-axis.
- **Click and drag:** Select a region to zoom into.
- **Double-click:** Reset the axes to show all data.

7.3 Dynamic Resampling (FigureResampler)

For high-frequency signals (e.g., waveforms sampled at hundreds of Hz), the application uses **Plotly-Resampler** to dynamically load detail as you zoom in. When viewing a long time range, the plot shows a downsampled overview. As you zoom into a shorter time window, the full-resolution data is loaded automatically.

This keeps the interface responsive even with millions of data points.

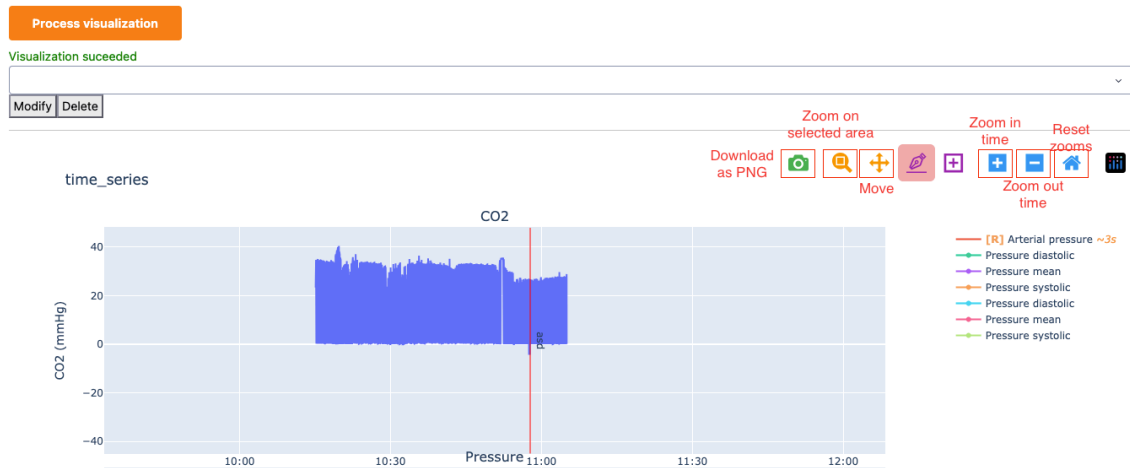


Figure 8: Interactive plot navigation

8 Annotations

The annotation system lets you mark events, time periods, or regions of interest directly on the plots. Annotations are saved to an `annotations.json` file in the `cdv_visu/` folder and persist across sessions.

8.1 Drawing Annotations

Use the Plotly drawing tools in each plot's toolbar:

- **Draw Line:** Click two points to draw a vertical line marking a specific event.
- **Draw Rectangle:** Click and drag to highlight a time region or value range.

Each annotations can be given a name and a color via the edit popup that appears after drawing.

8.2 Annotations Management

After processing, the **Annotation Controls** section becomes visible below the Process button:

- **Annotations Dropdown:** Lists all annotations across all figures, showing the annotations label.
- **Modify Button:** Opens the edit popup for the selected annotation (change name, color, or whether it spans all subplots).
- **Delete Button:** Removes the selected annotation.

8.3 Annotation Properties

Each annotation has the following properties:

- **Name:** A text label displayed on the annotation.
- **Color:** The line or fill color.
- **Global:** When enabled, the annotation spans all subplots in the figure (using paper y-coordinates).

8.4 Persistence

Annotations are automatically saved to `annotations.json` in the patient's `cdv_visu/` folder whenever you create, modify, or delete an annotation. They are reloaded when you re-process the same patient data.

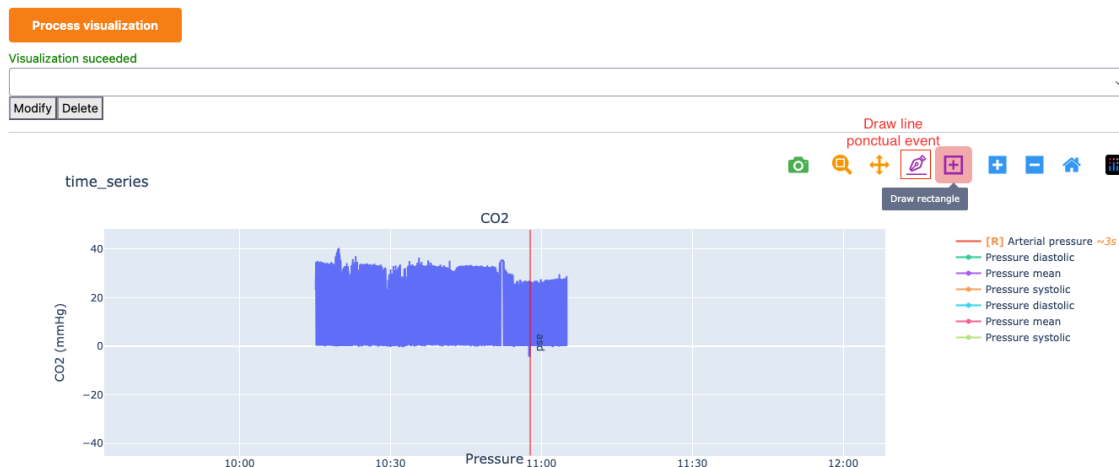


Figure 9: Annotations tools

9 Configuration File Reference

9.1 patient_options.json

This file defines patient-specific settings. It is automatically saved to the `cdv_visu/` subfolder each time you click “Process visualization”.

```
{
  "data_folder": "/path/to/patient/data",
  "datetime_start": "2024-10-08 10:00:00",
  "datetime_end": "2024-10-08 12:00:00",
  "quick_load": false,
  "philips_waves": {
    "time_shift": 20.0
  },
  "eit": {
    "day": "2024-10-08"
  }
}
```

Key	Type	Default	Description
<code>data_folder</code>	string	—	Path to the patient’s root data folder (required)
<code>datetime_start</code>	string or null	null	Start of the time window (YYYY-MM-DD HH:MM:SS). Leave empty to use all available data.
<code>datetime_end</code>	string or null	null	End of the time window. Leave empty to use all available data.
<code>quick_load</code>	boolean	false	Reuse previously cached .parquet files in <code>cdv_visu/</code>
<code><source_name></code>	object	—	Per-source options block (e.g., <code>time_shift</code> , <code>day</code>)

9.2 database_options.json

This file controls which data sources are active and how each signal is displayed. A snapshot is automatically saved to `cdv_visu/database_options.json` each time you click “Process visualization”.

9.2.1 Top-Level Structure

```
{
  "global": {
    "grouped_fields": { "Pressure": ["ART", "PNIs", "PNIm", "PNId"] }
  },
  "philips_waves": { ... },
  "philips_numerics": { ... },
  "eit": { ... }
}
```

Each data source key is optional — only include the sources you want to enable. The presence of a source key in this file is what activates that source; removing it disables it entirely.

9.2.2 Per-Source Block Structure

```
"philips_waves": {
  "field_display": ["ART", "PAP", "P-aer"],
  "signals": {
    "ART": {
      "label": "Arterial pressure",
      "unit": "mmHg",
      "unit_conversion": 1.0,
      "range": [-10, 200],
      "priority": 1.0,
      "color": "red",
      "visible": true,
      "line_dash": "solid",
      "period_resampling": 0.2,
      "hover_template": null
    }
  },
  "grouped_fields": {
    "Respiratory": ["P-aer", "CrbVol"]
  },
  "loop": {
    "pv_loop": ["P-aer", "CrbVol"]
  },
  "numerics": {
    "period_resampling": 0.5,
    "priority": 1.0
  },
  "additional_informations": {
    "timezone": "Europe/Paris"
  }
}
```

9.2.3 Per-Source Fields Reference

Key	Type	Default	Description
field_display	list of strings	all signals	Signal names to display. Signals absent from this list are loaded but hidden. Omit to show all.
signals	object	{}	Per-signal display options (see Per-Signal Fields Reference below).
grouped_fields	object	{}	Groups of signals to overlay on the same subplot, within this datasource. { "Respiratory waves": ["signal_1", "signal_2", ...], }

Key	Type	Default	Description
<code>loop</code>	object	<code>{}</code>	PV-loop definitions: <code>{"loop_name": ["x_signal", "y_signal"], ...}</code> .
<code>numerics</code>	object	<code>{}</code>	Datasource-level defaults for numeric parameters (see numerics Block below).
<code>additional_information</code>	object	<code>{}</code>	Device-level metadata, including timezone override (see additional_information Block below).

9.2.4 Per-Signal Fields Reference (`signals.<signal_name>`)

Key	Type	Default	Description
<code>label</code>	string	signal name	Display label shown on plot axes and legends.
<code>unit</code>	string	<code>" "</code>	Unit string shown on the Y-axis (e.g., <code>"mmHg"</code> , <code>"cmH2O"</code>).
<code>unit_conversion</code>	float	1.0	Multiplication factor applied to raw values for unit conversion.
<code>range</code>	<code>[min, max]</code> or null	auto	Fixed Y-axis range. Either bound can be null for auto-scaling.
<code>priority</code>	float	source default	Plot ordering priority (lower = higher on page). Signals with the same priority share a subplot.
<code>color</code>	string	auto	Line color (any CSS color string, e.g., <code>"red"</code> , <code>"#1f77b4"</code>).
<code>visible</code>	boolean	<code>true</code>	Set to <code>false</code> to load the signal but hide it from the plot by default.
<code>line_dash</code>	string	<code>"solid"</code>	Line style: <code>"solid"</code> , <code>"dash"</code> , <code>"dot"</code> , <code>"dashdot"</code> .
<code>period_resampling</code>	float	source default	Resampling period in seconds for this specific signal.

Key	Type	Default	Description
hover_template	string	null	Custom hover tooltip. Magic keywords: "fraction" shows values in (0, 1) as 1/n; "percentage" shows them as 33.3%. Any other string is passed directly to Plotly as a <code>hovertemplate</code> . Leave empty for the default compact display.

9.2.5 numerics Block (Datasource-Level Defaults)

The **numerics** block sets **default values** shared by all numeric parameter signals in a datasource — without listing each signal individually. Any per-signal entry inside **signals** takes precedence; signals without an explicit entry inherit from here.

```
"philips_numerics": {
  "numerics": {
    "period_resampling": 0.5,
    "priority": 2.0
  }
}
```

Key	Type	Default	Description
period_resampling	float	source default	Resampling period in seconds for all numeric parameters in this datasource.
priority	float	source default	Default plot priority for numerics (lower = higher on page). Overridden per signal by <code>signals.<name>.priority</code> .

This block functions as a datasource-wide default container for resampling and priority. It does not filter which signals are affected — it simply provides fallback values.

In the Excel format, these values are set via the **sentinel row** (`signal = *`). See [database_options.xlsx](#).

9.2.6 additional_informations Block

The **additional_informations** block carries device-level metadata that affects how raw data is interpreted. Currently its only field is **timezone**.

```
"eit": {
  "additional_informations": { "timezone": "UTC" }
}
```

Key	Type	Default	Description
<code>timezone</code>	string	source default	Override the timezone used to give a timezone (e.g., "Europe/Paris", "UTC") to timestamps which are timezone-naive. All plots still render in the configured display timezone.

Which datasources support the timezone override?

Datasource	Supports <code>timezone</code> override?
EIT, FluxMed Signals, FluxMed Parameters	Yes
Mindray Scope, Mindray Respi Waves, Mindray Respi Numerics	Yes
Syringe, Other (per-file, via <code>other::<stem>.additional_information.timezone</code>)	Yes
Philips Waves, Philips Numerics, Servo-U	No — timezone is fixed inside the source module

For datasources without override support, the key is silently ignored. The timezone is hardcoded in their source modules — this is a deliberate choice reflecting the vendor’s data format, not a library limitation.

In the Excel format, set the timezone via the **sentinel row** (`signal = *`, `timezone` column). See [database_options.xlsx](#).

9.2.7 Global Fields

```
"global": {
  "grouped_fields": {
    "Pressure": ["ART", "PNIs", "PNIm", "PNId"]
  },
  "loop": {
    "PV loop (ventilator vs. FluxMed)": [
      "servo_u:Paw", "fluxmed_signals:Vt"
    ]
  }
}
```

Key	Description
<code>global.grouped_fields</code>	Groups signals from different datasources onto the same subplot. Signal names must be unique across the datasources involved — use <code>datasource::raw_name</code> to disambiguate.
<code>global.loop</code>	Cross-datasource phase loops: <code>{"loop_name": ["x_ref", "y_ref"]}</code> . Both signals can come from different datasources. Each <code>*_ref</code> is resolved by the same three-mode chain as <code>grouped_fields</code> .

9.2.8 Signal Reference Resolution

Signal references in `grouped_fields` and `loop` — both global and per-source — are resolved by the following three-mode lookup chain:

1. **Qualified reference** `datasource::raw_name` — explicit and unambiguous. Recommended when the same column name exists in several datasources.
2. **Display name** — the `label` from the signals block. Works when the label is unique across sources; a warning is logged if it matches several signals.
3. **Raw name fallback** — the column name as it appears in the raw data file.

Multi-cycle loops are rendered with a **time-range slider** below the plot so you can scroll through cycles.

```
{ } example_database_options_other.json M X
example > option_files > { } example_database_options_other.json > { } other::waves > { } additional_informations

2  "global": {
3    "grouped_fields": {
4      "ART pressure (waves + numerics)": [
5        "other::waves::art",
6        "other::numerics::ARTm"
7      ]
8    },
9    "loop": {
10     "Other numerics vs Other mindray_respi_waves": [
11       "other::numerics::ARTm",
12       "other::mindray_respi_waves::MDC_PRESS_AWAY-MDC_DIM_CM_H2O"
13     ]
14   }
15 },
16
17 "other::waves": {
18   "field_display": ["art", "pleth"],
19   "additional_informations": {
20     "timezone": "Europe/Paris"
21   },
22   "signals": {
23     "art": {
24       "label": "Arterial Pressure",
25       "unit": "mmHg",
26       "color": "red",
27       "range": [0, 500],
28       "hover_template": "<b>ART:</b> %<{y:.1f}> mmHg<extra></extra>"
29     },
30     "pleth": {
31       "label": "Plethysmography",
32       "unit": "AU",
33       "color": "purple"
34     }
35   },
36   "grouped_fields": {
37     "Hemodynamic waves": ["art", "pleth"]
38   },
39   "loop": {
40     "Art vs Pleth (within-file)": ["art", "pleth"]
41   }
42 },
43
44
45 "syringe": {
46   "field_display": [
47     "Vitesse Propofol",
48     "Vitesse Remifentanyl"
```

Figure 10: example json database options file

9.3 database_options.xlsx

For clinical users, the **Excel format is the recommended way** to configure signals — it requires no knowledge of JSON and can be edited in any spreadsheet application. On upload, it is automatically converted to the equivalent [JSON structure](#), so every option available in JSON is also available in the spreadsheet.

The file must contain a sheet named **signals** and optionally a sheet named **loops**.

9.3.1 signals sheet

One row per signal. The columns **datasource** and **signal** are mandatory; all others are optional and fall back to the defaults listed in the per-signal table above.

Use ***** in the **signal** column to write a **sentinel row** that sets datasource-level defaults (e.g., a common **period_resampling** or **timezone**) without defining a specific signal — equivalent to the [numerics](#) and [additional_informations blocks](#) in JSON. Column names are case-insensitive (e.g., **Label**, **UNIT**, **Hover_Template** all work).

The **Scope** column below indicates where each field is meaningful:

- **Both** — valid in sentinel (*) and per-signal rows
- **Signal** — per-signal rows only; ignored in sentinel rows
- **Sentinel** — sentinel (*) rows only; a warning is logged if set in a per-signal row

Column	Required	Scope	Description
datasource	Yes	Both	Data source name (e.g., philips_waves , eit).
signal	Yes	Both	Raw signal name. Use * for a sentinel row that sets datasource-level defaults.
label	No	Signal	Display label. Defaults to the signal name if empty or identical.
unit	No	Signal	Unit string (e.g., mmHg).
unit_conversion	No	Signal	Numeric multiplier for unit conversion.
range_min	No	Signal	Minimum Y-axis value.
range_max	No	Signal	Maximum Y-axis value.
priority	No	Both	Plot priority (float). In a sentinel row sets the datasource-level default; in a signal row overrides it for that signal only.
color	No	Signal	CSS color string.
visible	No	Signal	yes / no (default: yes). Accepts yes , 1 , true , oui , vrai (case-insensitive).
line_dash	No	Signal	solid , dash , dot , dashdot .

Column	Required	Scope	Description
<code>period_resampling</code>	No	Both	Resampling period in seconds. In a sentinel row sets the datasource-level default; in a signal row overrides it for that signal only.
<code>hover_template</code>	No	Signal	Hover tooltip format. Magic keywords: <code>"fraction"</code> shows values in (0, 1) as 1/n; <code>"percentage"</code> shows them as 33.3%. Any other string is forwarded directly to Plotly as a <code>hovertemplate</code> .
<code>display</code>	No	Signal	<code>yes / no</code> — whether to add this signal to the display list. Default: <code>yes</code> .
<code>groups</code>	No	Signal	Semicolon-separated group names (e.g., <code>Respiratory;Pressure</code>). Groups within one datasource become local <code>grouped_fields</code> ; groups spanning multiple datasources become <code>global.grouped_fields</code> .
<code>timezone</code>	No	Sentinel	Override the timezone for this datasource (e.g., <code>"Europe/Paris"</code> , <code>"UTC"</code>). Only valid in <code>*</code> rows; a warning is logged if placed in a per-signal row. Works with <code>other::<stem></code> datasource keys. See additional_informations Block for which datasources support this.

9.3.2 loops sheet (optional)

One row per PV-loop definition — equivalent to the `loop` key in the [JSON per-source block](#) or in `global`. If the sheet is absent or malformed it is silently skipped.

Column	Required	Description
<code>datasource</code>	Yes	Data source that owns both signals.
<code>loop_name</code>	Yes	Name for the loop plot (e.g., <code>p_v_loop</code>).
<code>x_signal</code>	Yes	Signal name for the X axis.

Column	Required	Description
y_signal	Yes	Signal name for the Y axis.

See [example/option_files/](#) in the source repository for complete example files in both formats.

<

10 Troubleshooting

10.1 Browser Does Not Open Automatically

If the browser does not open after launching the application, manually navigate to:

`http://127.0.0.1:8050`

Ensure no other application is using port 8050 (typically a previous app launch terminal tab not yet closed). If needed, close the terminal window which was opened in the app and restart the application.

10.2 No Data Found

If the visualization is empty or a data source shows no signals:

- Start by inspecting the data (teal button) rather than visualizing it, which provides much more information on what the app could gather from the data.
- Try with default visualization database options, to ensure it's not your options that hides the data from being visualized
- Verify that the **data folder path** is correct and accessible.
- Check that subfolders follow the **naming conventions** (see Section 3).
- Ensure the subfolder contains files with one of the **accepted extensions** for that data source (see Section 3). Files with unrecognized extensions are silently ignored.
- For **single-file** sources: if the folder contains multiple unrelated data files (different stems), the source is skipped. Keep only one data file per folder, or provide the same data in multiple formats (e.g., `data.csv` + `data.parquet`) and the preferred format will be selected automatically.

10.3 Slow Loading

Large datasets may take time to load on the first run. To speed up subsequent loads:

- Enable the “**Re-use data if already loaded once**” (`quick_load`) option. This uses the cached `.parquet` files in `cdv_visu/` instead of re-reading raw data files.

10.4 Time Alignment Issues

If signals from different sources appear misaligned in time:

- Use the **Time shift** option in the per-source settings to adjust alignment.
- Verify that the correct **day** or **date** is set for sources that require it (e.g., EIT).

10.5 Application Crashes or Errors

- Check the terminal window for error messages.
- Log files are available in the `logs/` directory.
- Ensure the data files are not corrupted or truncated.
- If you think you are facing a real bug, please report it. (to the author `@alexis.janin@inria.fr` or on `gitlab/github`)

10.6 Known limitation

These may or may not be tackled in the future, depending on the needs of the users. Feel free to ask for one of the below or any other feature demand/bug report on the `gitlab/github` page.

- No timeshift inside a datasource, e.g. if 2 timeseries from `philips_waves` are not aligned, this currently can't be solved in the app.
- Display timezone in plots is hardcoded to Europe/Paris. This will be modified.